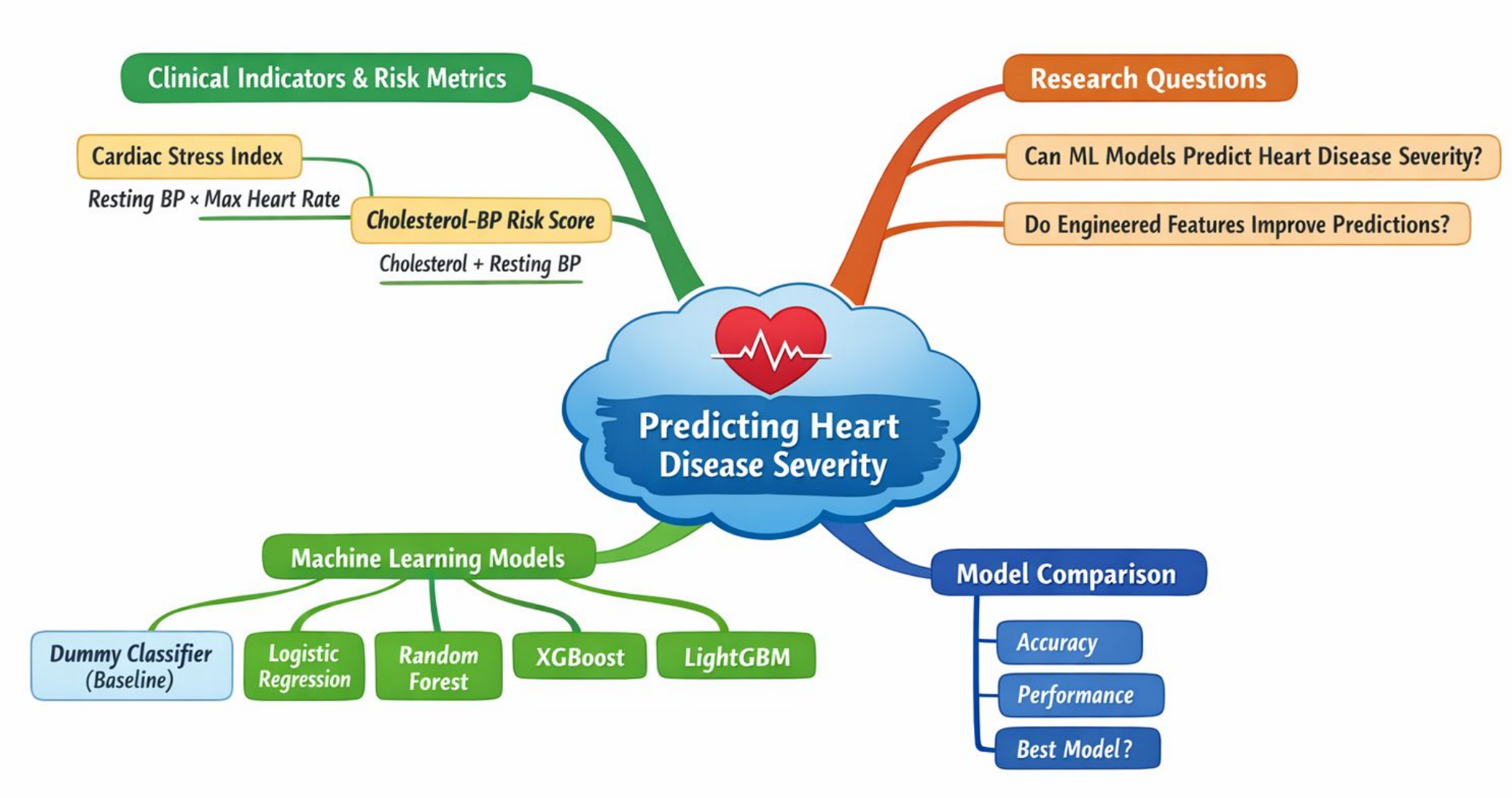
National Canadian Medical Datathon 2026

Predicting Heart Disease Severity Using Clinical Indicators and Derived Cardiovascular Risk Metrics: A Comparison of Five Machine Learning Models

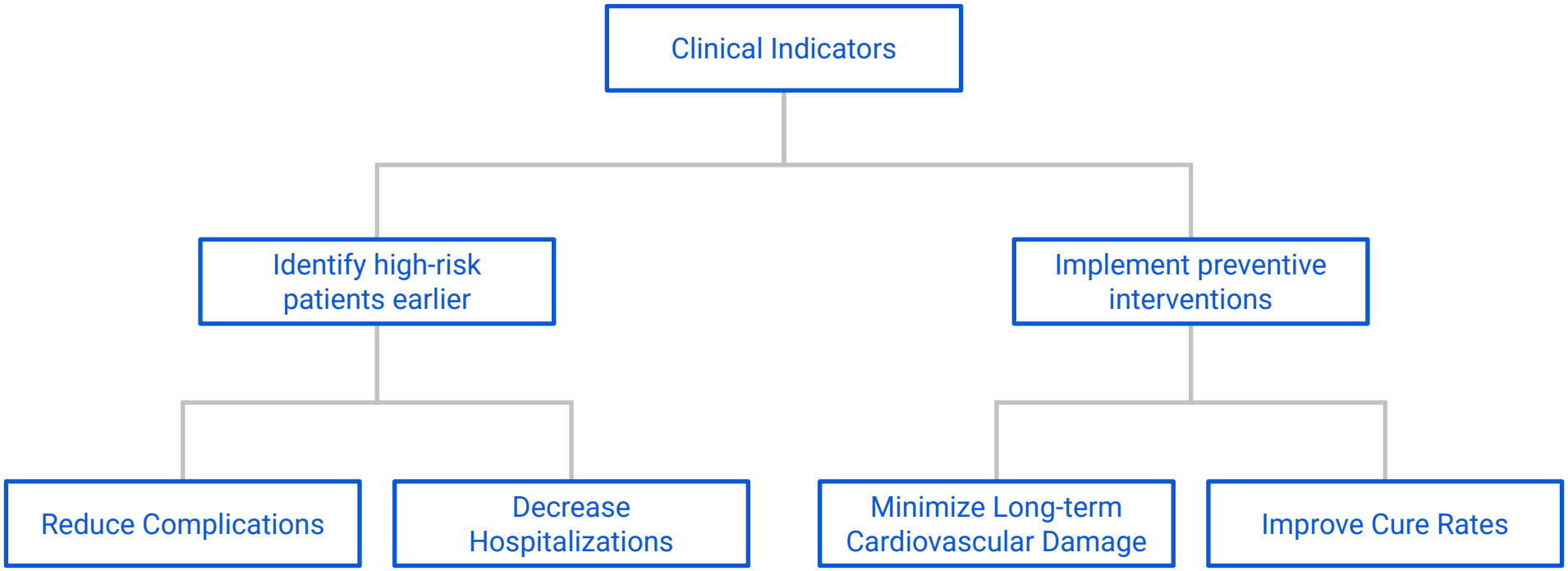
Team 9

March 8, 2026

Problem Definition

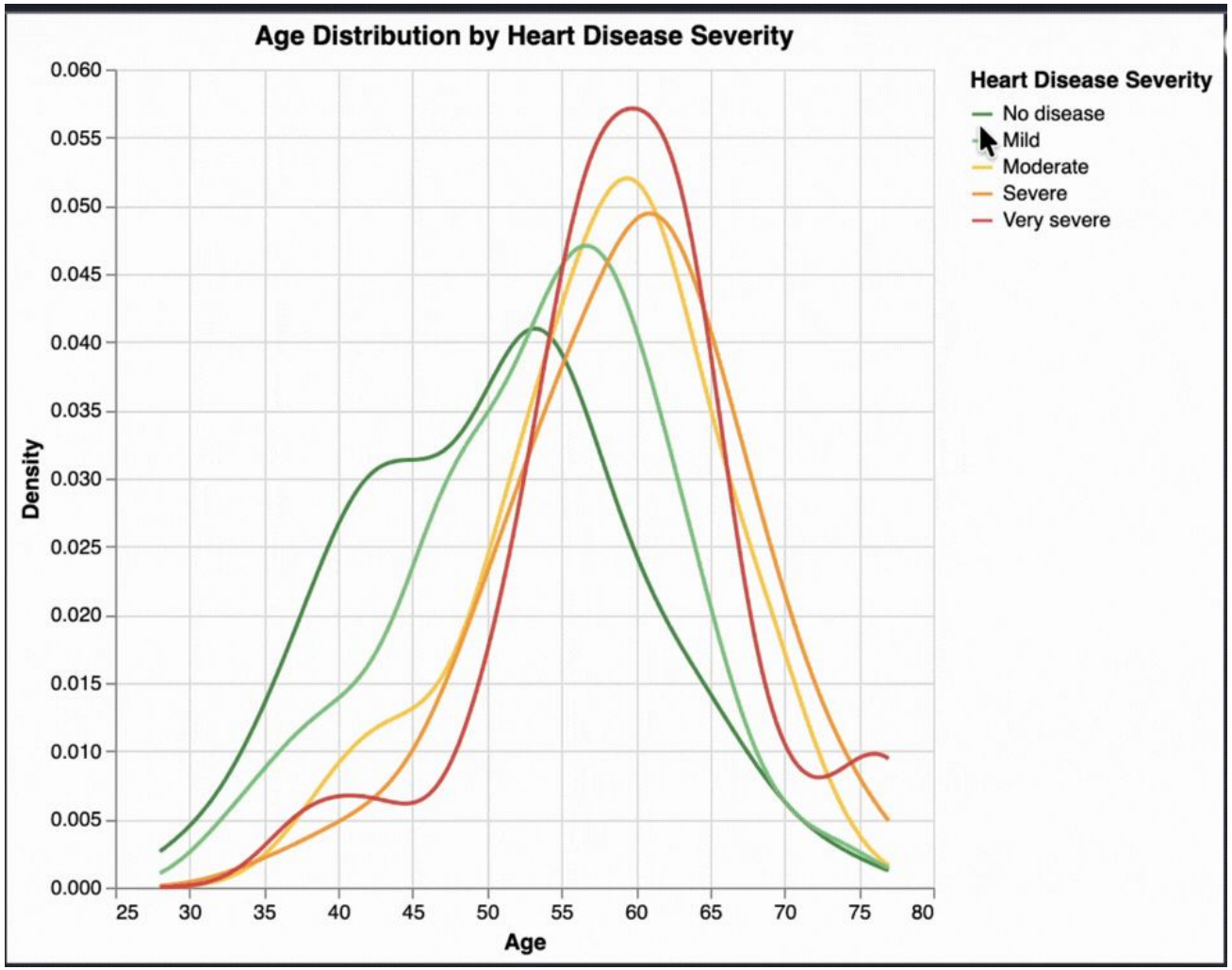


Clinical Relevance

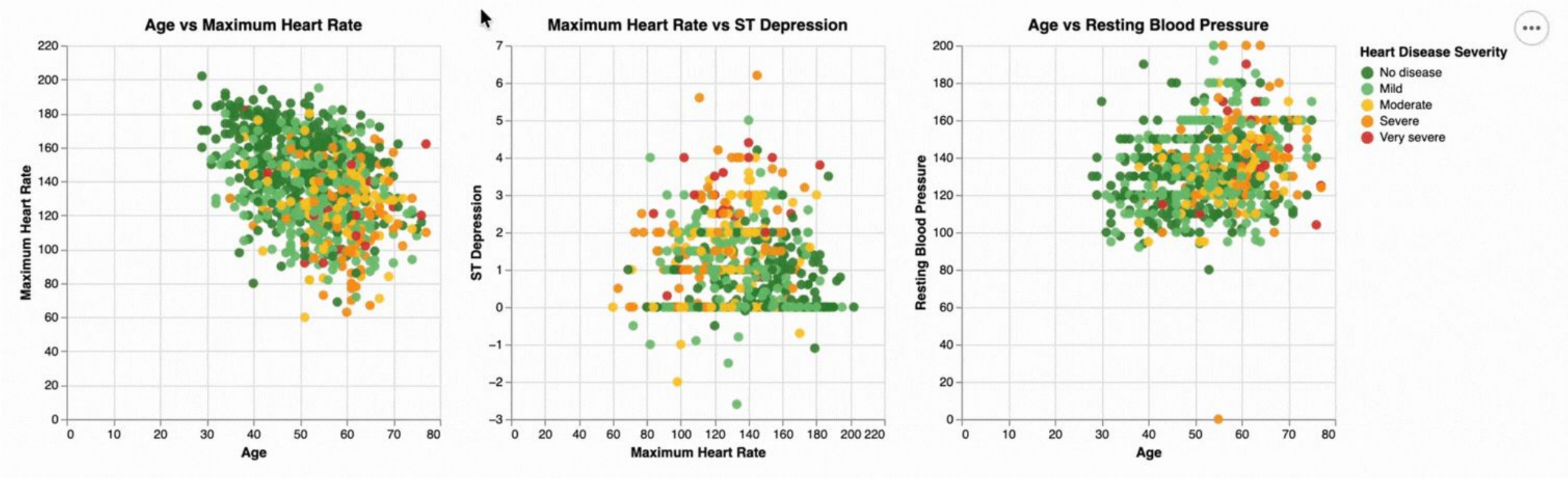


Visualization

- Compare how age shifts as heart disease becomes more severe.
- Age may be an important risk factor that controls heart disease progression.
- But age alone cannot fully separate severity levels.



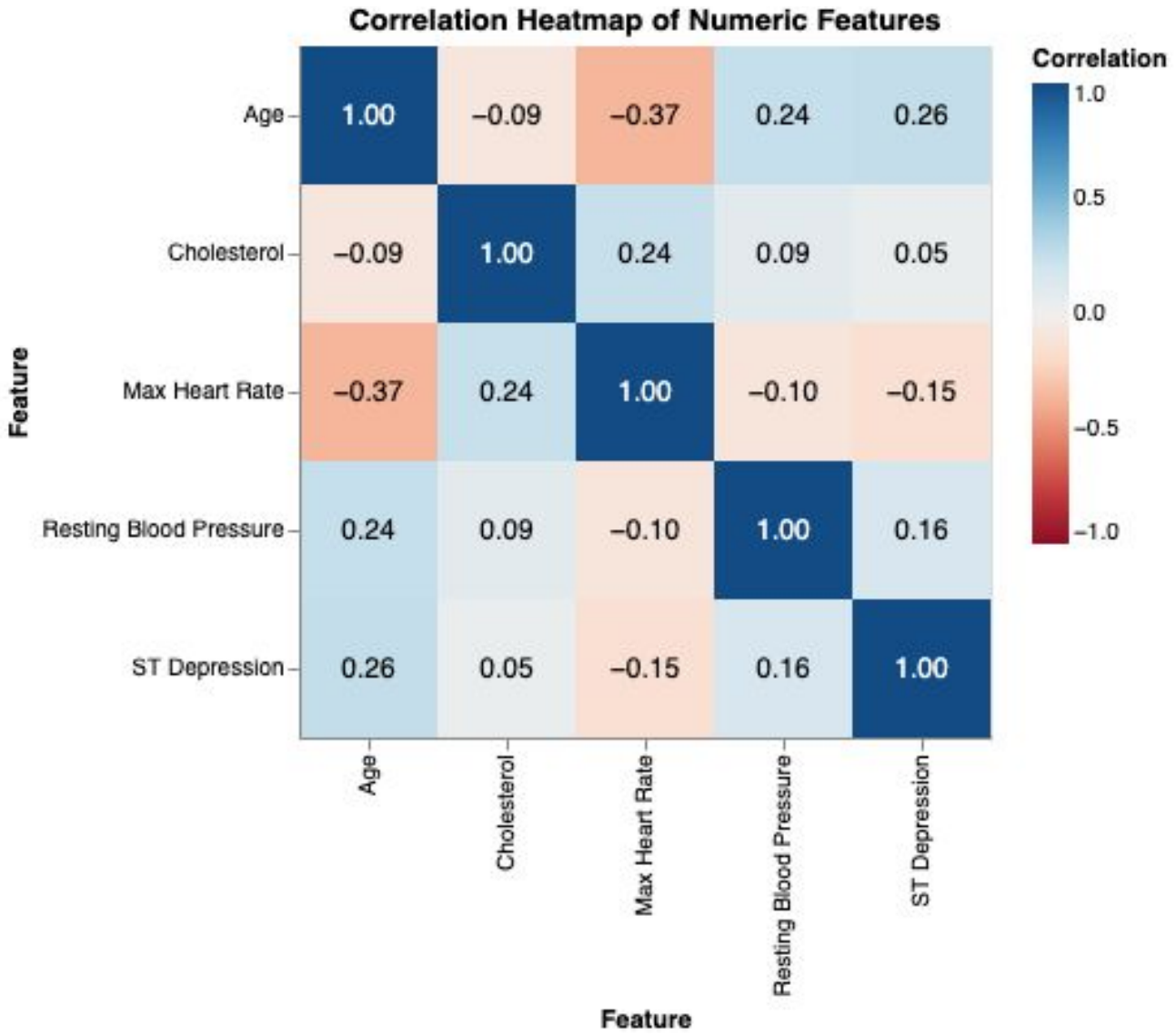
Visualization



- Maximum heart rate decreases as age increases (negative relationship).
- Severe cases appear more often at higher ages and lower maximum heart rates.
- Higher ST depression values are more common among severe cases.

Visualization

- Most variables have weak correlations → suggesting low multicollinearity;
- The strongest relationship is a negative correlation between age and maximum heart rate (−0.37);
- There is also a mild positive correlation between age and resting blood pressure (0.24).



Data Understanding & Preprocessing

Numeric	Categorical	Binary
Missing values: Imputation: Mean	Missing values: imputed with the most frequent value	Missing values: imputed with the most frequent value
Scaling: StandardScaler	**Encoding:** One-Hot Encoding	**Encoding:** One-Hot (drop if binary)

Data Understanding & Preprocessing

Derived Metric	Formula	Clinical Interpretation
Cardiac Stress Index	`trestbps × thalach`	Combines **resting blood pressure** and **maximum heart rate** to estimate cardiac workload. Higher values may indicate greater cardiovascular stress during activity.
Cholesterol–Blood Pressure Risk Score	`chol + trestbps`	Combines **cholesterol level** and **blood pressure** to capture combined cardiovascular risk, since both factors are strongly associated with heart disease.

Scoring metrics

Class Imbalance

- The dataset is **imbalanced across classes**
- Labels represent disease severity:

  - **0** – No disease

  - **1** – Mild disease

  - **2–4** – Increasing levels of severe disease

Limitations of Accuracy

- Accuracy may be **misleading with imbalanced data**

Why Macro F1

- **F1 score balances precision and recall**
- **Macro averaging treats each class equally**
- Provides a **fair evaluation across all disease severity levels**

Analysis & Methodology

Baseline Model

- Used **Dummy Classifier** (predicts most frequent label)
- Test Recall: **0.36,** Macro F1: **0.34**
- Serves as a **baseline for model improvement**

Linear Model (Logistic Regression)

- Test Recall improved to **0.663**
- Macro F1 improved to **0.639**
- Shows that **basic linear relationships already capture useful patterns**

Non-Linear Models

- Tested three tree-based models:
  - **Random Forest**
  - **LightGBM**
  - **XGBoost**

- **Random Forest** achieved the best initial performance among untuned models

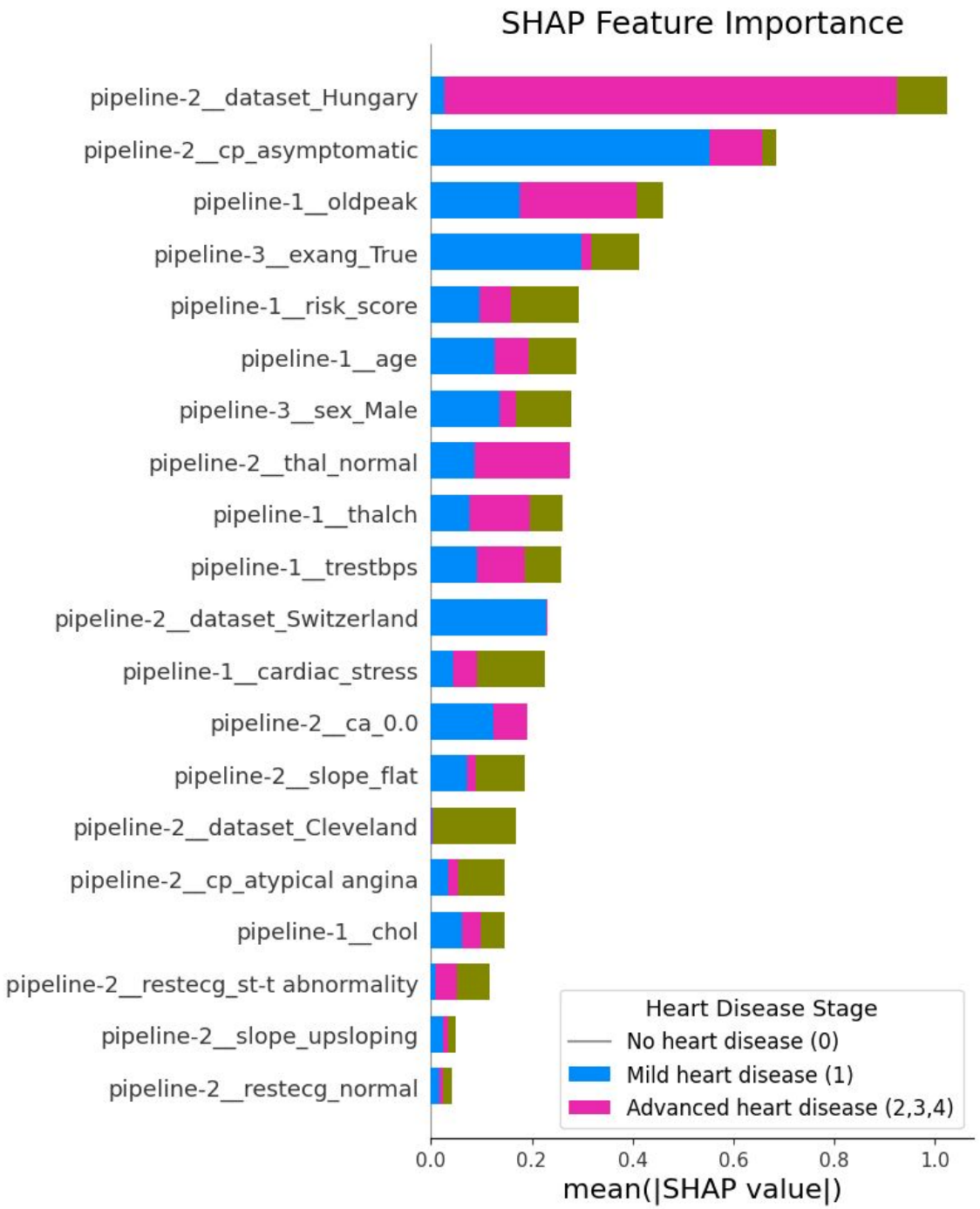
  - Test Accuracy ≈ **0.686**
  - Macro F1 ≈ **0.650**

Hyperparameter Tuning

- Performed tuning to optimize model parameters
- Improved model generalization and performance

Final Model Choice

- **Tuned LightGBM** selected as the final model
- Achieved the **best overall balance** of:
  - Accuracy
  - Weighted Recall
  - Macro F1
- Performance improved by **~0.30 compared to baseline**

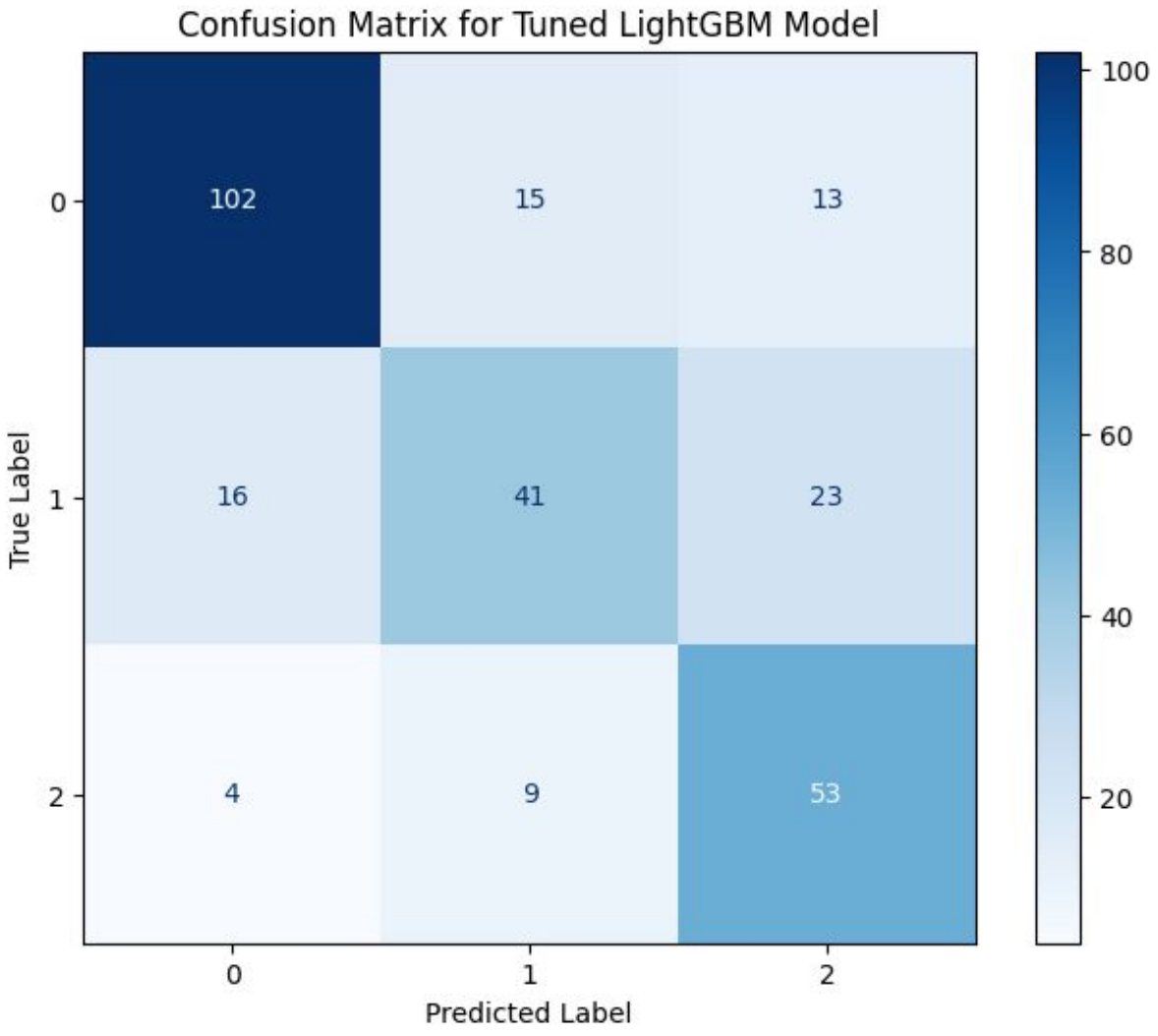


Three important features:

- **Dataset origin - Hungary**
- **Clinical feature - Asymptomatic chest pain**
- **Electrocardiogram measure - Oldpeak**

- LightGBM correctly classifies most **healthy** and **severe** cases.

- **Moderate** cases are harder to distinguish.



Key Findings & Practical Implications

Clinical Value

1

Early screening for high-risk patients

Supports early detection and risk stratification

2

Clinical decision support systems

Helps predict heart disease severity more accurately

3

More efficient allocation of diagnostic resources

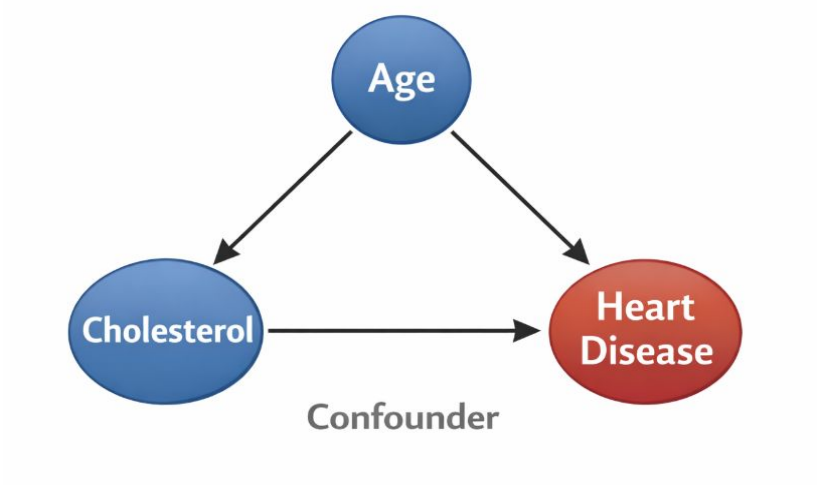
Focus on reducing false negatives

Model Limitation

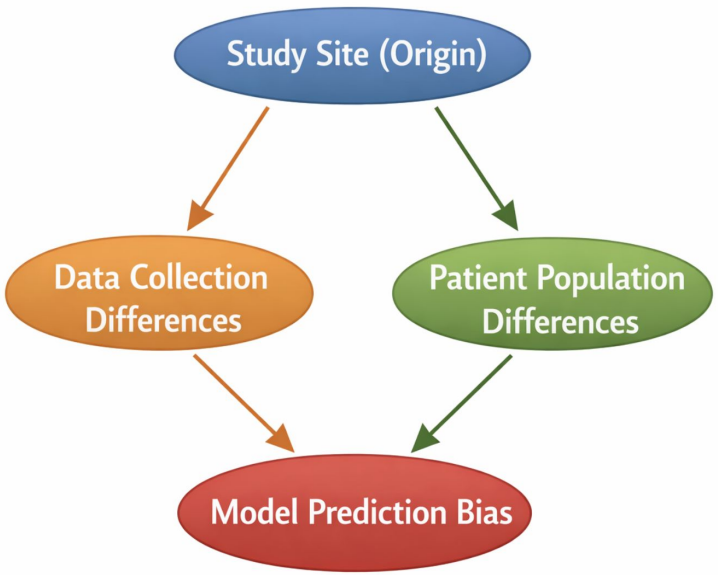
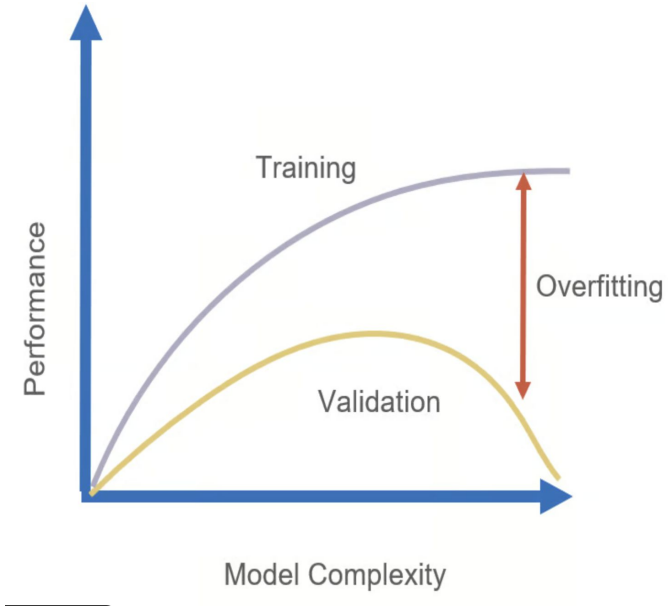
Confounders

Cholesterol → Heart Disease

Age → Cholesterol
Age → Heart Disease



Representative



Overfitting